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SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	30 June, 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality of the Smart Agricultural Production Optimization Engine (OptiCrop*)* project. The system is developed using Machine Learning and Flask to provide intelligent crop recommendations and environmental suitability analysis.

The application processes agricultural parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to generate accurate predictions. The code is modular, well-documented, follows Python coding standards, and satisfies the functional requirements of the system.

Solution Summary

Field	**Details**
Repository Link / URL	keerthipriya03/OptiCrop_Smart_Agricultural_Production_Optimization_Engine
Programming Language(s)	Python 3.11, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Pickle
Key Features Implemented	• Data preprocessing and cleaning • Feature handling for soil nutrients (N, P, K) and environmental factors • Model training using Machine Learning algorithms (Decision Tree, Random Forest) • Best model saved using Pickle • Flask-based web application for real-time prediction • User-friendly input form for applicant details • Instant prediction of suitable crop based on input data
Pending / Incomplete Features	• User login authentication • Database integration for storing prediction history • Email/SMS notification system • API integration for real-time weather data
Setup / Run Instructions	1. Install required libraries: pip install -r requirements.txt 2. Run the application: python app.py. 3. Open a web browser and navigate to: `http://127.0.0.1:5000` 4. Enter soil and environmental parameters and click **Predict** to view crop recommendation

Code Quality Checklist

S.No	Criteria	Status(yes/no)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The OptiCrop project follows good software engineering practices by maintaining a modular structure, reusable components, and clear documentation. Machine learning models are trained and evaluated to select the most effective algorithm for crop recommendation.
The integration of the trained model with a Flask web application enables real-time predictions based on user input. The system can be further enhanced by incorporating